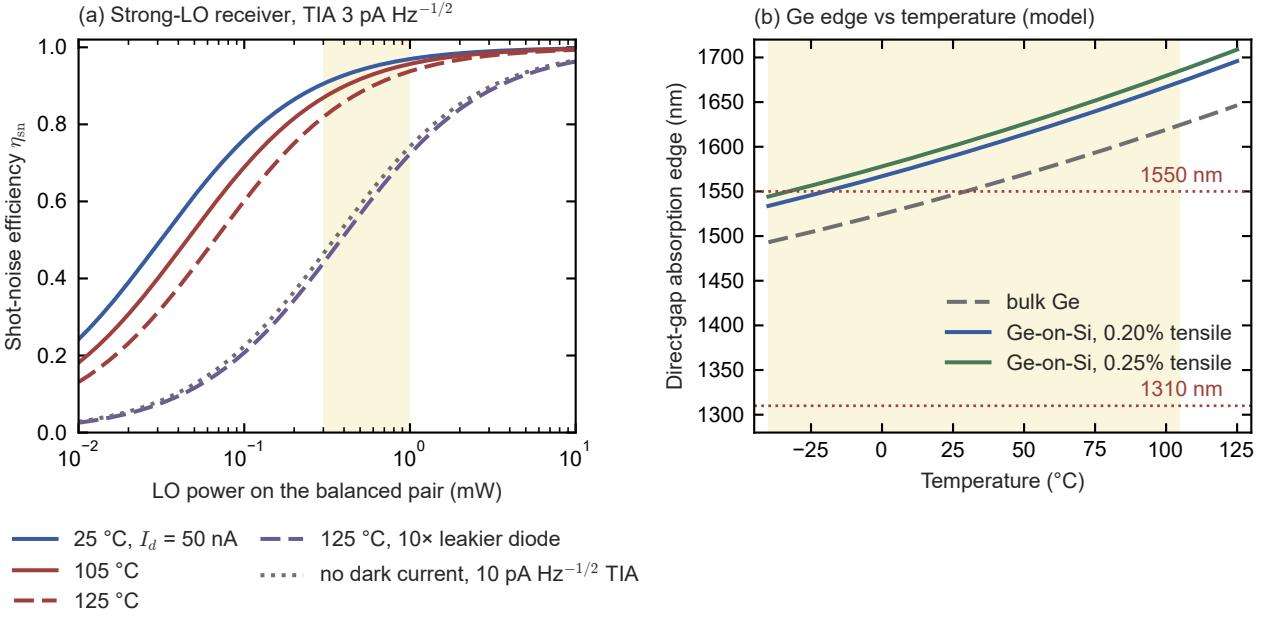
(a) Strong-LO receiver, TIA 3 pA Hz$^{-1/2}$
Shot-noise efficiency η_{sn}
1.0
0.8
0.6
0.4
0.2
0.0
10^{-2}
10^{-1}
10^{0}
10^{1}
LO power on the balanced pair (mW)
25 °C, I_d = 50 nA
105 °C
125 °C
125 °C, 10× leakier diode
no dark current, 10 pA Hz$^{-1/2}$ TIA
(b) Ge edge vs temperature (model)
Direct-gap absorption edge (nm)
1700
1650
1600
1550
1500
1450
1400
1350
1300
1550 nm
1310 nm
bulk Ge
Ge-on-Si, 0.20% tensile
Ge-on-Si, 0.25% tensile
−25
0
25
50
75
100
125
Temperature (°C)